

RAG Knowledge Base Guide

Semantic Document Ingestion, Vector Chunking, and Analytics Grounding

One Janitorial Operations Library

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1. Overview & Purpose

Technical reference manual for managing the Retrieval-Augmented Generation (RAG) platform. Explains chunking and vector lookup.

Details document upload rules, vector spacing configurations, and search parameters.

Explains how vector indexing is connected to the analytics platform.

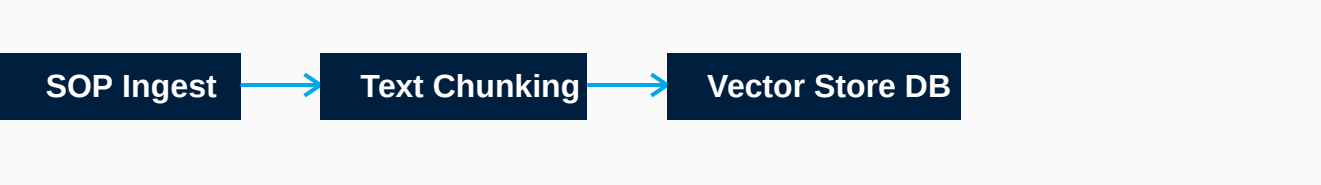
2. Key Features & Blueprint

Document Chunking Processor: Logic dividing texts into 2000-character blocks with 200-character overlaps.

Vector Search Engine: Pipeline retrieving context blocks matching user input.

RAG Grounding: Pipeline matching customer complaints to specific training chunks.

3. System Process Flow



4. Step-by-Step Usage & Operations

Ingesting new SOPs: Navigate to RAG section, select document, run validation, and click Index Document.

Querying indexed context: Navigate to SOP query panel, type search keys, and view retrieved chunks.

Evaluating Search Weights: Run sample queries inside tests box, and audit returned relevance factors.

5. Best Practices & Safety Guidelines

- Keep text files clean of non-standard characters to optimize chunk divisions.
- Set overlap thresholds appropriately to maintain contextual continuity across sections.
- Index updated manuals promptly to ensure chatbot answers remain correct.

6. Troubleshooting & FAQ

- Vector search returns empty blocks: Confirm target file is fully indexed and check database connection.

Parser fails: Ensure file encoding is UTF-8 before loading into the ingestion dashboard.

Irrelevant results returned: Increase relevance threshold values inside search options.

7. Glossary & Appendix

RAG: Retrieval-Augmented Generation - Grounding AI output in files.

Chunking: Splitting documents into small text chunks.

Cosine Similarity: Metric calculating semantic closeness.